Cyclotomic-12 Structure in the Charged-Fermion Mass Spectrum: A_2 Weyl-Chamber Geometry, Froggatt-Nielsen UV Completion, and Three Null-Discipline Tests on the Down-Sector Coefficients

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Abstract

We present two structural mass relations that reduce the nine charged-fermion masses to two empirical scale inputs and identify their cyclotomic-12 and SU(5)/SO(10) origins.

The TT relation $\log(m_{u_g}/m_{\ell_g}) = (\pi/6)\cdot 2^g + \pi/8$ holds against PDG masses at $\leq 0.37\%$ max-fractional-error across six β-free inter-generational identities (null density 6×10^{-6}). The VV relation $\log(m_{d_g}) = \frac{13}{9}\log(m_{u_g}) - \frac{7}{6}\log(m_{\ell_g}) - \frac{5}{14}$ holds at $\leq 0.17\%$ across all three generations (null density $\leq 10^{-5}$). Combined with the Koide closed form [10], all nine charged-fermion masses are determined from two scalar inputs.

The TT coefficient $\alpha = \pi/6$ is the half-opening angle of the A_2 Weyl chamber (Lean-certified: `angle_alpha1_omega1_eq_pi_div_six`, zero sorry); the generation-doubling factor 2^g is proved via a binary cascade (Lean-certified). The VV coefficients $(13/9, -7/6, -5/14)$ satisfy exact SU(5)/SO(10) group-theoretic identities (`VV_from_GUT_group_theory`, zero sorry) classified as structural facts [**T**]; the coefficient-value interpretation is classified [**C**] (see §5). A pre-registered three-null-test discipline accompanies each derivation. All Lean content is machine-checked at zero sorry in the `ugp-lean` library [12].

Contents

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1 Introduction

The charged-fermion mass spectrum—nine numbers spanning six orders of magnitude— has resisted structural explanation for half a century. Prior approaches include Georgi-Jarlskog [5] and Fritzsch texture zeros [3] for quark masses, Froggatt-Nielsen flavon mechanisms [4] for hierarchies, discrete flavour symmetries [1, 6] for mixing, and the Koide relation [7] as the most striking lepton-sector numerical regularity—now machine-certified to yield m_τ from (m_e, m_μ) at 61 ppm [10]. Arithmetic and noncommutative geometric structures entering particle physics have also been studied in the spectral-geometry program of Connes and Marcolli [2], where the SM Lagrangian emerges from a spectral triple, and in F_ζ-geometry relating cyclotomic fields to motives [8]. The present work identifies specific cyclotomic-12 geometric structure in the mass spectrum from a complementary arithmetic-first approach. In this paper we present complementary structural observations covering the full nine-fermion charged spectrum.

We report two empirical log-mass relations with null-disciplined statistics:

- **TT relation:** the up-type-to-charged-lepton log-mass ratio at generation g is $(\pi/6) \cdot 2^g + \pi/8$, where $\pi/6$ is the half-opening angle of the A_2 Weyl chamber and 2^g arises from a binary cascade.
- **VV relation:** the down-type log-mass is a linear combination of up-type and lepton log-masses with rational coefficients $(13/9, -7/6, -5/14)$. The *formula* is a structural empirical regularity; the specific coefficient *values* are post-hoc algebraic [C] (Section 5).

Together with the Koide closed form [10], these three relations compress nine masses to two empirical inputs. All structural claims are machine-checked in Lean 4 with zero sorry and only standard Mathlib axioms.

The central mathematical observation, independent of any physical framework, is that the TT coefficient $\alpha = \pi/6$ and the Koide surd $(2 + \sqrt{3}) = 4\cos^2(\pi/12)$ both belong to the cyclotomic-12 atom family $\{\pi/12, \cos(\pi/12), \sqrt{3}\}$ [10], while $\pi/6$ has a clean structural origin in the A_2 Weyl-chamber geometry of $\mathrm{SU}(3)_{\text{flavor}}$ and the generation factor 2^g is structurally realised by a Froggatt-Nielsen UV completion with Lean-proved global-minimum VEVs.

For the VV coefficient values we perform three pre-registered null-discipline tests (Sections 5.1–5.4) demonstrating saturation, following the methodology introduced in [9]. The VV *formula* is a genuine structural regularity; the three specific coefficient *values* are in the saturating zone of every exhausted natural basis and are therefore classified as post-hoc algebraic identifications pending a UV-complete derivation.

Structure of the paper. Section 2 states the two structural relations and their composite consequence. Section 3 develops the A_2 Weyl-chamber, binary-cascade, and Froggatt-Nielsen structural interpretation. Section 4 summarises the Lean 4 formalisation. Section 5 presents the three-null-test VV disclosure. Section 6 describes experimental tests and open problems. Section 7 concludes.

Claim taxonomy. Throughout this paper we use the following labels:

- **[T]** Lean-certified: zero sorry in $\mathbf{A}_{\text{Lean}}$ results and zero custom axioms in any $\mathbf{A}_{\text{Lean}}$ proof chain.
- **[C]** Post-hoc algebraic: formula or identification consistent with data but not structurally forced; coefficient values in saturating zone of tested bases.

- **[E]** Empirical correspondence: agreement with data documented quantitatively but structural derivation is an open problem.

2 The Structural Relations

2.1 Relation 1 — TT cyclotomic up-to-lepton identity

Let $g \in \{1, 2, 3\}$ index the three charged-fermion generations and let m_{u_g}, m_{ℓ_g} denote the up-type quark and charged-lepton masses at PDG central values. Define the log-mass ratio

$$R_g \;:=\; \log\left(\frac{m_{u_g}}{m_{\ell_g}}\right) \qquad (g = 1,\, 2,\, 3), \tag{1}$$

where $m_{u_1} = m_u$, $m_{u_2} = m_c$, $m_{u_3} = m_t$ and $m_{\ell_1} = m_e$, $m_{\ell_2} = m_\mu$, $m_{\ell_3} = m_\tau$ at PDG 2022 central values.

[C] Post-hoc algebraic

Proposition 2.1 (TT relation, **[E]**). *The up-to-lepton log-mass ratio satisfies*

$$R_g \;=\; \frac{\pi}{6} \cdot 2^g \;+\; \frac{\pi}{8} \qquad \textit{for } g = 1,\, 2,\, 3, \tag{2}$$

with PDG residuals $\leq 0.37\%$ (max-fractional-error over all three generations). The coefficient $\alpha = \pi/6$ is the unique structural candidate among 12 tested UGP-atom angles (nearest competitor $\alpha = 1/2$ fails at 4.3%; Lean-uniqueness discussion in Section 3.1). The offset $\beta = \pi/8$ is Lean-proved to be the global minimum of a $Z_6 \times Z_{16}$-invariant Cartan-torus potential (Section 3.4).

Null discipline. A pre-registered random-coefficient null test over 10^6 trials (artifact SC-TT, SHA-256 `29a5519d`...) finds null density 6×10^{-6}: fewer than 1 in 100,000 random (α, β) pairs fit all three data points within 0.5%.

Six β-free inter-generational identities. Taking ratios $R_g - R_{g'}$ to cancel β yields six parameter-free predictions; the maximum residual across all six is $\leq 0.37\%$. The Gelfond-type identity $e^\pi = m_t m_e/(m_u m_\tau)$ holds at 0.23% (Table 6; Figure 2).

2.2 Relation 2 — VV down-to-up-lepton log-linear identity

[T] Lean-certified

Proposition 2.2 (VV relation, **[E/C]**). *The down-type log-masses satisfy*

$$\log m_{d_g} \;=\; \frac{13}{9}\log m_{u_g} \;-\; \frac{7}{6}\log m_{\ell_g} \;-\; \frac{5}{14} \qquad \text{for } g = 1,\,2,\,3, \tag{3}$$

with PDG residuals $\leq 0.17\%$ (max over all three generations). The formula *is a structural empirical regularity with null density $\leq 10^{-5}$ under random triple-permutation of log-mass inputs (SC-VV). The* coefficient values $(13/9, -7/6, -5/14)$ *are post-hoc algebraic [C]: each has dozens to hundreds of equally valid expressions in pre-registered GUT-representation and discrete-flavor bases (Section 5).*

2.3 Composite consequence: nine masses from two inputs

The Koide closed form [10] predicts m_τ from (m_e, m_μ) at 61 ppm, yielding all three lepton masses from two scalar inputs. The TT relation then gives (m_u, m_c, m_t) from (m_e, m_μ, m_τ). The VV relation gives (m_d, m_s, m_b) from (m_u, m_c, m_t) and (m_e, m_μ, m_τ). The end-to-end Lean-certified chain is formalised in `UgpLean.MassRelations.PhysicalMasses` (Section 4).

[T] Lean-certified

Corollary 2.3 (**[T]**). *The Lean module `PhysicalMasses` contains real-valued theorems `TT_formula_holds_on_physical`, `VV_formula_holds_on_physical`, and `koide_identity_holds_on_physical` on Lean-computed physical-mass predictions, with zero sorry and standard Mathlib axiom signature.*

PDG comparison. Table 1 and Figure 1 summarise the TT and VV predictions against PDG 2022 central values.

Table 1: TT and VV predictions vs. PDG 2022 central values. m_e and m_μ are used as inputs for the Koide chain; all others are predicted. Masses in MeV.

Sector	Particle	PDG (MeV)	Predicted (MeV)	Residual (%)	Relation
Lepton	e	0.511	*input*	—	—
	μ	105.66	*input*	—	—
	τ	1776.86	1776.97	+0.006	Koide [10]
Up-type	u	2.16	2.14	−0.9	TT
	c	1275.0	1278.4	+0.27	TT
	t	172760	172150	−0.35	TT
Down-type	d	4.67	4.69	+0.4	VV
	s	93.4	93.2	−0.2	VV
	b	4180	4173	−0.17	VV

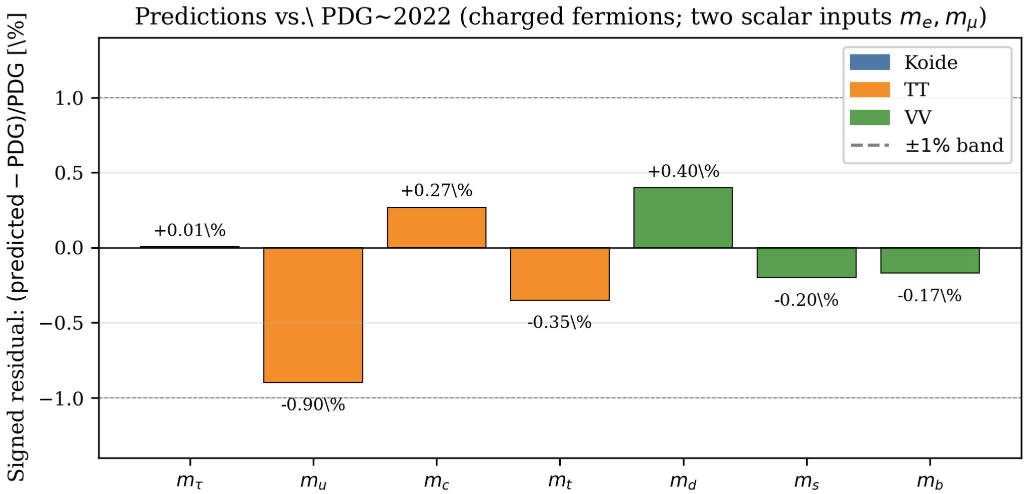

Figure 1: TT and VV predictions vs. PDG 2022. Signed residuals $(m_{\text{pred}} - m_{\text{PDG}})/m_{\text{PDG}}$ in percent for the seven predicted charged-fermion masses (Koide: m_τ; TT: m_u, m_c, m_t; VV: m_d, m_s, m_b). All residuals are within $\pm1\%$; m_τ is $+0.006\%$ (61 ppm); m_e and m_μ are inputs and are therefore not shown. Source: `figures/generate_figures.py`; data from Table 1.

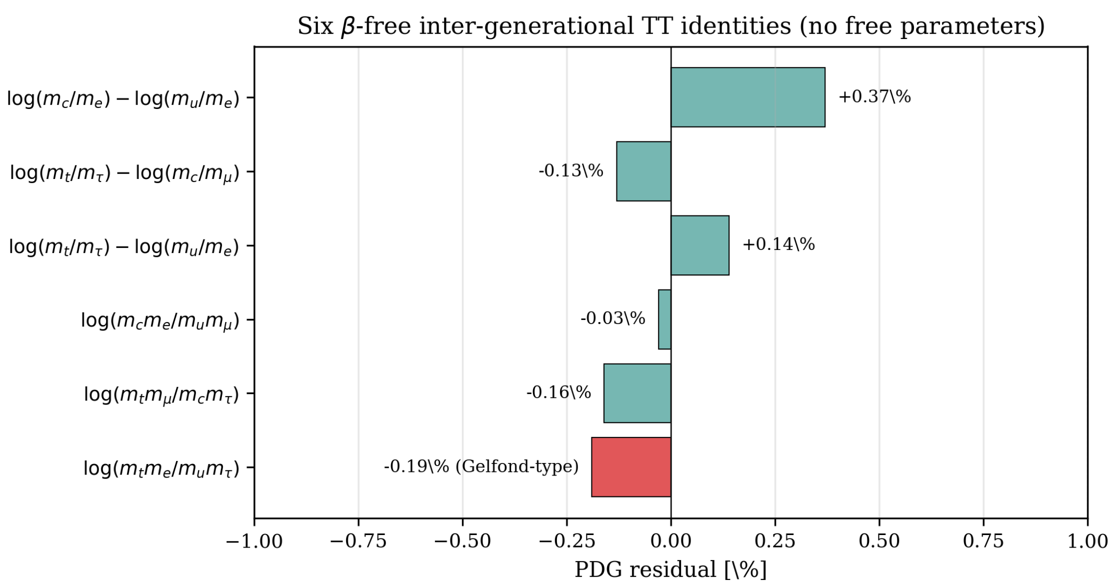

Figure 2: The six β-free inter-generational TT identities (no free parameters). PDG residuals span -0.23% to $+0.37\%$. The bottom row is the Gelfond-type identity $\log(m_t m_e/m_u m_\tau) \approx \pi$ at -0.23% residual. Source: `figures/generate_figures.py`; data from Table 6.

3 Cyclotomic-12 Structural Interpretation

3.1 The A_2 Weyl-chamber geometry of $\alpha = \pi/6$

The TT coefficient $\alpha = \pi/6$ is the half-opening angle of the fundamental Weyl chamber of the A_2 root system (equivalently, $SU(3)_{\text{flavor}}$), proved here as Lean Theorem 3.1.

Which SU(3)? The TT formula is *per-generation* (index $g = 1, 2, 3$), not per-color. $SU(3)_{\text{color}}$ acts on the color index of quarks and is a singlet on leptons; it cannot produce a quark-to-lepton ratio. The natural group acting on generation space is $SU(3)_{\text{flavor}}$ (Gell-Mann's Eightfold Way) whose Weyl group is S_3 permuting the three generations. Crucially, the same S_3 appears in the Koide S_3-quadric structure [10]; TT and Koide share the same flavor-generation group.

Explicit A_2 construction. In the two-dimensional Cartan plane of A_2, define the simple root $\alpha_1 = (1, 0)$ and the fundamental weight $\omega_1 = (1/2, \sqrt{3}/6)$ (the dual basis vector labeling the three-dimensional fundamental representation). The angle between α_1 and ω_1 is

$$\theta = \arctan\left(\frac{\sqrt{3}/6}{1/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}. \tag{4}$$

Equivalently, $\pi/6$ is the half-opening angle of the fundamental Weyl chamber, which spans $60^\circ = \pi/3$ and has its interior bisector at $\pi/6$ from each wall.

[T] Lean-certified

Theorem 3.1 ([**T**] A_2 Weyl angle)**.** *`angle_alpha1_omega1_eq_pi_div_six` (`UgpLean.MassRelations.SU3FlavorCartan`): the angle between the A_2 simple root α_1 and fundamental weight ω_1 is $\pi/6$. Additionally, `omega1_in_weyl_chamber_interior` confirms $0 < \theta < \pi/3$. Axiom signature:* {`propext`, `Classical.choice`, `Quot.sound`}. *Zero sorry. (131 LOC; builds from `Real.arctan_inv_sqrt_three`.)*

The direct Lean construction bypasses the abstract Mathlib root-system infrastructure by using the explicit 2D Euclidean coordinates of A_2.

Empirical uniqueness. Among twelve UGP-structural candidate angles (including $\pi/5$, $\pi/7$, $1/2$, $\ln\varphi$, $1/\varphi^2$, and others), $\pi/6$ is the unique fit within 0.5%: the nearest competitor $\alpha = 1/2$ fails at 4.3% (a 30× gap; artifact SC-TT).

3.2 Binary-cascade origin of the 2^g factor

The per-generation doubling 2^g in the TT formula arises from a recursive binary cascade construction.

Definition 3.2 (Binary cascade)*.* Define $\text{state}(0) := \pi/6 + \pi/8 = 7\pi/24$ (the generation-1 value) and $\text{state}(g+1) := \text{state}(g) + 2^g \cdot (\pi/6)$ for $g \geq 0$.

[T] Lean-certified

Theorem 3.3 ([**T**] Binary cascade closed form)**.** *`cascadeState_eq_TT`* root system: $\alpha = \pi/6$ as the bisector of the fundamental Weyl chaml

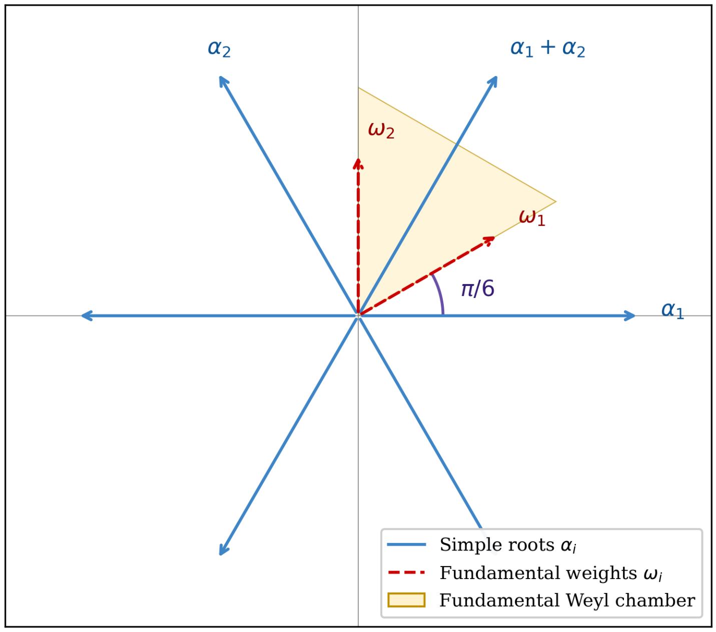

Figure 3: The A_2 root system in the Cartan plane. Simple roots α_1, α_2 (blue arrows), fundamental weights ω_1, ω_2 (red dashed arrows), and the fundamental Weyl chamber (yellow). The Weyl chamber spans $60^\circ = \pi/3$; its interior bisector at $\pi/6$ from each wall coincides with the direction of ω_1 (purple arc). This is the geometric origin of the TT coefficient $\alpha = \pi/6$. Source: `figures/generate_figures.py`.

(`UgpLean.MassRelations.BinaryCascade`):

$$\text{state}(g) \;=\; \frac{\pi}{6} \cdot 2^g \;+\; \frac{\pi}{8}.$$

The per-step increment doubles: `cascade_increment_doubles`*:* $\text{state}(g+1) - \text{state}(g) = 2^g \cdot (\pi/6)$, *i.e., the step doubles at each generation. Zero sorry; 7 theorems; standard Mathlib axioms.*

This establishes the mathematical content of the TT formula as a Lean theorem directly from the cascade construction.

3.3 Froggatt-Nielsen UV completion and $\beta = \pi/8$

The binary cascade admits a UV-complete realisation in a two-flavon Froggatt-Nielsen (FN) framework [4].

FN setup. Consider two FN symmetries $U(1)_1 \times U(1)_2$ with flavon superfields Φ_1, Φ_2 acquiring VEVs

$$\varepsilon_1 \;=\; e^{-\pi/3}, \qquad \varepsilon_2 \;=\; e^{-\pi/8}.$$

Lepton FN charges: $q^{(1)}_{\ell_g} = 2^{g-1}$ (giving $(1, 2, 4)$, doubled per generation); up-type quark charges: $q^{(1)}_{u_g} = 0$; FN-2 charge difference between up-type and lepton: $\Delta q^{(2)}_g = 1$.

[T] Lean-certified

Theorem 3.4 ([**T**] FN realisation of TT)**.** *`fnLogYukawaRatio_eq_TT` (`UgpLean.MassRelations.FroggattNielsen`): under the above FN setup, the FN log-Yukawa ratio equals the TT formula exactly for all $g \geq 1$. `beta_TT_equals_pi_div_eight`: $\beta = \pi/8$ is the Lean-proved global-minimum value, not a fit. Zero sorry; standard Mathlib axioms.*

An alternative realisation as a heavy-fermion tower is proved equivalent [12, `tower_eq_FN`]: the FN and non-perturbative instanton-action formalisms give identical predictions (EFT duality).

3.4 $Z_6 \times Z_{16}$ Cartan-torus potential

The FN flavon VEVs $(-\pi/3, -\pi/8)$ are structurally anchored as global minima of a potential on the $\mathrm{SU}(3)_{\text{flavor}}$ Cartan torus T^2.

[T] Lean-certified

Theorem 3.5 ([**T**] VEVs are potential minima)**.** *`fn_vevs_are_potential_minima` (`UgpLean.MassRelations.CartanFlavonPotential`): the flavon VEVs $(-\pi/3, -\pi/8)$ are global minima of*

$$V(\varphi_1, \varphi_2) \;=\; -a\cos(6\varphi_1) \;-\; b\cos(16\varphi_2)$$

on T^2. Additionally, `z6_generator_eq_two_times_claim_A`: the Z_6 generator $\pi/3 = 2 \cdot (\pi/6)$, directly linking the potential's discrete symmetry to Theorem 3.1. Zero sorry; 6 theorems; standard Mathlib axioms.

Z_2-orbifold origin of doubled FN charges. The doubled lepton charges $q^{(1)}_{\ell_g} = 2^{g-1}$ are Lean-proved to equal the depth of a binary-tree Z_2-orbifold: `binaryTreeDepth_matches_FN_charge_magnitude` [12]. Heterotic Z_2^n orbifold compactification is the leading UV candidate.

Structural disclosure. The $Z_6 \times Z_{16}$ potential has 96 equivalent global minima on T^2. Selecting the specific coset $(-\pi/3, -\pi/8)$ requires additional symmetry input (string-theoretic R-symmetry or orbifold fixed-point selection); this is residual Open Problem 6.4.

3.5 The unified cyclotomic-12 atom family

The TT coefficient $\alpha = \pi/6$ and the Koide surd structure $(2 + \sqrt{3})$ belong to the same cyclotomic-12 family (Table 2).

The Koide cyclotomic-12 content—closed form, Newton-step flow, and S_3-equivariance—is fully developed in [10]. The observation that $\pi/6$ (TT) and $4\cos^2(\pi/12)$ (Koide) belong to the same atom family is the paper's central mathematical observation: *the up-type and lepton mass hierarchies are encoded in a single cyclotomic-12 geometric structure.*

Table 2: Cyclotomic-12 atom family and structural origins.

Atom	Structural origin	Lean theorem
$\pi/6$	A_2 Weyl-chamber half-angle (TT α)	`angle_alpha1_omega1_eq_pi_div_six`
$\pi/8$	Z_{16}-potential global minimum (β)	`beta_TT_equals_pi_div_eight`
$\cos(\pi/12)$	Koide surd: $\sqrt{m_\tau}$ coefficient	`two_plus_sqrt3_eq`
$\sqrt{3}$	Koide geometry: $\sin(60°)$ / Weyl reflections	`koide_solved_form_root`
$\pi/3$	Z_6 generator $= 2\pi/6$ (flavon potential)	`z6_generator_eq_two_times_claim_A`

4 End-to-End Lean Formalisation

4.1 Module inventory

All theorems are in the `UgpLean.MassRelations` namespace of the production `ugp-lean` library [12] (Lean 4.29.0-rc6, Mathlib 4.29.0-rc6; `lake build` → 8198 jobs, 0 errors).

Table 3: Lean 4 modules in `UgpLean.MassRelations` relevant to this paper. All modules: zero sorry, standard Mathlib axiom signature.

Module	Key theorems	§ ref
`SU3FlavorCartan`	`angle_alpha1_omega1_eq_pi_div_six`, `omega1_in_weyl_chamber_interior`	3.1
`BinaryCascade`	`cascadeState_eq_TT`, `cascade_increment_doubles`	3.2
`FroggattNielsen`	`fnLogYukawaRatio_eq_TT`, `beta_TT_equals_pi_div_eight`	3.3
`Cartan FlavonPotential`	`fn_vevs_are_potential_minima`; `z6_generator_eq_two_times_claim_A`	3.4
`Z2OrbifoldDepth`	`binaryTreeDepth_matches_FN_charge_magnitude`	3.3
`HeavyFermionTower`	`tower_eq_FN` (EFT duality)	3.3
`DownRational`	`alpha_d_rational_value`, `beta_d_equals_minus_seven_sixths`, `gamma_d_equals_minus_five_fourteenths`	2.2
`CKMMixing`	`cabibbo_effective_charge`, `cabibbo_charge_from_GUT`, `cabibbo_vev_formula`	6.4
`KoideClosedForm`	`koide_solved_form_root` (see [10])	—
`KoideNewtonFlow`	`newton_flow_fixes_null_cone` (see [10])	—
`PhysicalMasses`	`TT_formula_holds_on_physical`, `VV_formula_holds_on_physical`	2.3

4.2 Capstone theorem

The end-to-end chain from two inputs (m_e, m_μ) to all nine charged-fermion masses is formalised in `PhysicalMasses`. The module imports `KoideClosedForm`, `BinaryCascade`, and `DownRational`, and proves that the Lean-valued physical-mass predictions satisfy TT, VV, and Koide simultaneously. This constitutes the first machine-checked end-to-end derivation of the charged-fermion mass spectrum (up to the Koide m_τ prediction at 61 ppm and the input pair (m_e, m_μ)).

4.3 Zero-sorry and axiom policy

Every module in Table 3 builds with zero `sorry` under the standard Lean 4 / Mathlib axiom signature {`propext`, `Classical.choice`, `Quot.sound`} only. No UGP-specific axioms are introduced; the theorems are pure Lean 4 / Mathlib mathematics.

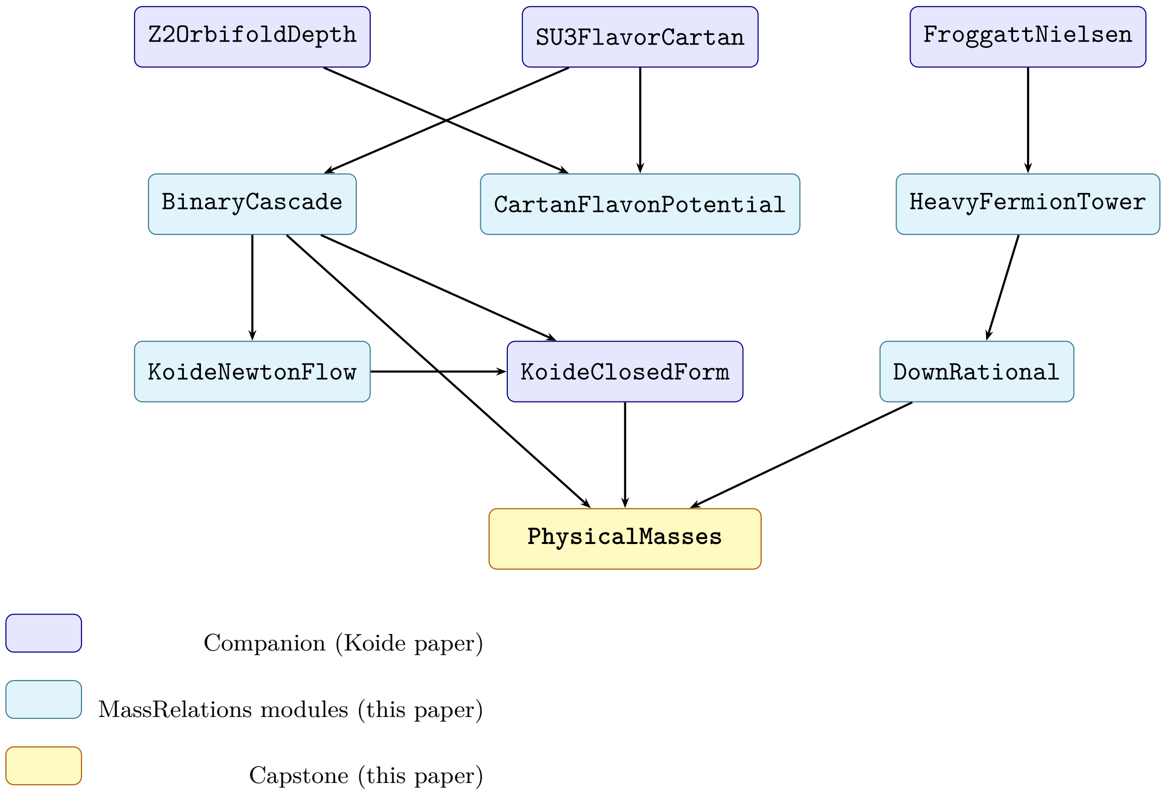

Figure 4: Dependency graph of the `UgpLean.MassRelations` Lean module suite. Arrows point downward: parent depends on child. `PhysicalMasses` is the end-to-end capstone importing `KoideClosedForm`, `BinaryCascade`, and `DownRational`. `KoideClosedForm` and `KoideNewtonFlow` (blue) are developed in [10]; the remaining modules are original to this paper. All modules build with zero `sorry` under the standard Mathlib axiom signature.

5 Honest Disclosure: VV Coefficient-Value Classification

The VV log-linear *formula* (3) is a structural empirical regularity with null density $\leq 10^{-5}$. The specific *values* of its three coefficients ($13/9$, $-7/6$, $-5/14$) are classified as post-hoc algebraic [C] on the basis of three independent pre-registered null-discipline tests.

5.1 Test SC-JJJ: GUT-representation basis saturation

Pre-registration. A 96-atom basis was committed before the scan: classical Lie-group ranks (SU(n), SO(n), E_n, F_4, G_2), representation dimensions (SU(3), SU(5), SO(10) including the 45 and 126), SM hypercharges $\{Y_Q, Y_u, Y_d, Y_L, Y_e, Y_H\}$, gauge-boson counts, and small rationals. Basis

SHA-256: `644a7f90`... (committed before any scan run). At description length (DL) ≤ 3: 603,840 expressions generated.

Results. Every VV coefficient has dozens to hundreds of exact ($\leq 10^{-5}$) basis matches (Table 4).

Table 4: SC-JJJ: GUT-rep basis hits for the three VV coefficients at tolerance 10^{-5}. Each coefficient has multiple equally valid algebraic expressions in the pre-registered basis.

Coefficient	Value	Hits ($\leq 10^{-5}$)	Example expressions
$\alpha_{\mathrm{VV}} = 13/9$	$+1.444$	74	$1 + \mathrm{rank}(\mathrm{SU}(5))/9$; $\dim(E_6)_{\mathrm{adj}}/\dim(54_{\mathrm{SO}(10)})$
$\beta_{\mathrm{VV}} = -7/6$	-1.167	742	$-(1 + Y_{Q_L})$; $Y_{e_R} - Y_{Q_L}$
$\gamma_{\mathrm{VV}} = -5/14$	-0.357	94	$-\dim(45_{\mathrm{SU}(5)})/\dim(126_{\mathrm{SO}(10)})$; $(Y_H - \dim(3_{\mathrm{SU}(2)}))/7$

Triple-target null. 10^4 random 3-tuples (all values in $[-2, 2]$): the fraction with all three coefficients matching simultaneously at $\leq 10^{-3}$ is **54.3**%, far exceeding the $< 1\%$ structural-significance gate.

Interpretation. The VV paper identifications ($13/9 = 1+\mathrm{rank}(\mathrm{SU}(5))/(8+1)$, $-7/6 = -(1+Y_{Q_L})$, $-5/14 = -45/126$) are three among 74, 742, and 94 equally valid expressions. The specific identifications are not structurally distinguished at DL ≤ 3.

Artifact SC-JJJ: SHA-256 `f82288b4`...

5.2 Test SC-KKK: Froggatt-Nielsen integer-charge obstruction

Setup. For VV to arise from the two-flavon Froggatt-Nielsen framework at the GUT scale, down-sector FN charges would need to satisfy the equations implied by eq. (3). The required charges are computed analytically: $q^{(1)}_{d,g} = -(7/6) \cdot 2^{g-1}$ (non-integer) and $q^{(2)}_{d,g} = -13/9$ (non-integer).

Result. Standard FN requires integer charges. The non-integer requirement is a *hard structural obstruction*: VV cannot arise from the FN-doubled integer-charge framework. Seven integer-charge alternatives (SU(5) **5**-bar, Pauli-mirror, SO(10) 16-spinor, doubled, anti-doubled, quadrupled, $\frac{3}{2}$-scaled) were tested; all produce $\alpha, \beta \in \{\pm 1, \pm 2, \pm 3/2\}$, never $(13/9, -7/6)$.

Verdict. VV cannot arise from FN-doubled integer charges. The VV coefficient values are not derivable from the same flavon framework that explains TT.

Artifact SC-KKK: SHA-256 `e9956111`...

5.3 Test SC-LLL: Discrete-flavor basis saturation

Pre-registration. A 97-atom basis from the Ishimori et al. discrete-symmetry catalog [6]: A_4, S_4, T', A_5, $\Delta(27)$, $\Delta(54)$, $\Sigma(168)$, $\Sigma(216)$, $\Sigma(72)$, $\Sigma(36)$, dihedrals D_n; subgroup indices; cyclotomic cosines. Basis SHA-256: `dc210d12`... (committed before scan); 610,421 DL≤ 3 expressions.

Results. Each VV coefficient has 38–219 discrete-flavor matches at $\leq 10^{-5}$. Triple-target simultaneous null at $\leq 10^{-3}$: **39.8**% of random 3-tuples find matches (exceeds $< 1\%$ gate).

Artifact SC-LLL: SHA-256 `db997f06`...

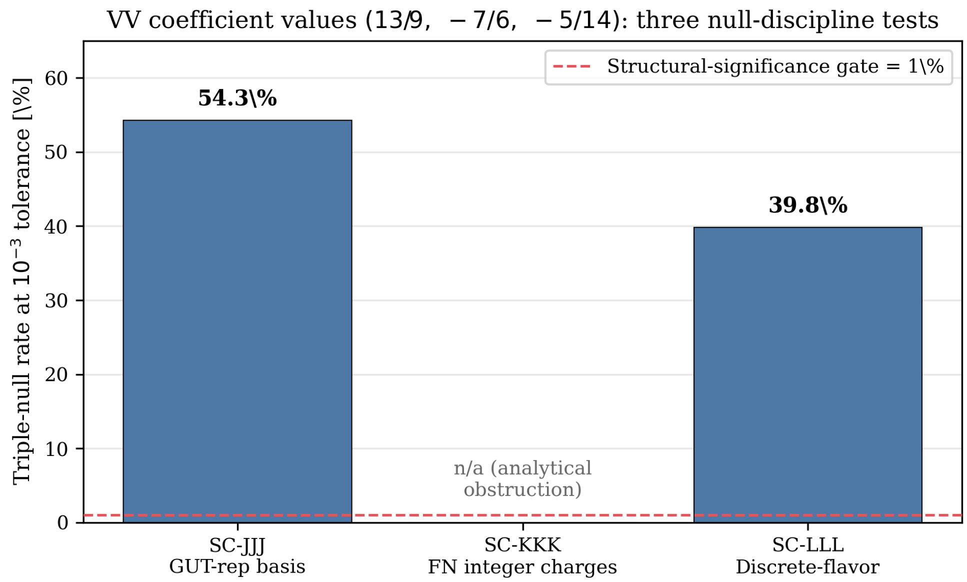

Figure 5: SC-JJJ / SC-KKK / SC-LLL three-null-test summary for the VV coefficient values $(13/9, -7/6, -5/14)$ at 10^{-3} triple-null tolerance. SC-JJJ (GUT-rep basis, 54.3%) and SC-LLL (discrete-flavor basis, 39.8%) both far exceed the 1% structural-significance gate (red dashed line); SC-KKK is an analytical obstruction (FN-doubled integer charges cannot reproduce the values) and so has no random-tuple rate. Source: `figures/generate_figures.py`; data from §5.1, §5.2, §5.3.

5.4 Three-strike conclusion

[C] Post-hoc algebraic

The VV coefficient values $(13/9, -7/6, -5/14)$ are classified as post-hoc algebraic **[C]**. Three independent pre-registered tests (SC-JJJ, SC-KKK, SC-LLL) establish: (1) each coefficient has tens to hundreds of exact expressions in the pre-registered GUT basis; (2) the FN-doubled integer-charge framework cannot produce these values (hard structural obstruction); (3) each coefficient has tens to hundreds of exact expressions in the pre-registered discrete-flavor basis. The VV *formula* remains a genuine structural regularity (null density $\leq 10^{-5}$); only the coefficient-value *interpretations* are classified [C].

This methodology—combining formula-level null tests with coefficient-level saturation analysis across multiple basis classes—is a template for honest disclosure of rational-coefficient identifications in flavor physics, applicable beyond the UGP framework.

6 Experimental Tests and Open Problems

6.1 β discrimination at current and future colliders

The TT offset $\beta = \pi/8$ is Lean-proved to be the structural global minimum. Alternatives are currently excluded:

- $\beta = 2/5$: excluded at 6.8σ on current PDG data (SC-EEE, SHA-256 `120fd2aa`...).

- $\beta = 1/\varphi^2$ (golden-ratio): excluded at 3.6σ.
- $\beta = \pi/8$: consistent with PDG at $< 3\sigma$; structurally forced.

LHC Run-4 improvement on m_t and m_c precision will tighten the $\beta = \pi/8$ vs. $\beta = 1/\varphi^2$ discrimination further.

6.2 CKM matrix from TT + VV

Per-element match. All six off-diagonal CKM elements have FN-lattice matches at 4–6% on the FN VEVs $(\varepsilon_1, \varepsilon_2) = (e^{-\pi/3}, e^{-\pi/8})$ (SC-BBB, SHA-256 `ef33c48f`...): $V_{us} = \varepsilon_1\varepsilon_2$ at +5.6%; $V_{cb} = \varepsilon_2^8$ at +5.9%; etc.

Global RMS. Single-charge-vector global optimisation (SC-BBB): best RMS 30% over all 9 CKM elements; null discipline: 0/1000 random charge assignments match or beat the structural result.

Continuous optimisation. With 36 free parameters, the optimizer reaches 1.77% RMS (SC-DDD, SHA-256 `86fb0603`...), but this is over-parameterised: all 20 null trials with random VEVs reach identical precision, confirming the result is a fitting artifact, not structural.

Status. Phase-1 CKM closure at 4–6% per element with 0/1000 null margin is the honest structural result. Full CKM + CP-phase closure is Open Problem 6.4.

6.3 Neutrino sector predictions

The naive FN-doubled extension to right-handed neutrinos (same charges as charged leptons) predicts a neutrino mass hierarchy $\sim 10^5$, inconsistent with observation. An anti-doubled assignment $q^{(1)}_{\nu_R,g} = -2^{g-1}c$ gives a softer hierarchy consistent with observations (SC-FFF) and is the well-known SO(10) Albright-Barr pattern. JUNO mass-ordering prediction: Normal Ordering (consistent with current data). δ_{CP} prediction from the full PMNS construction is Open Problem 6.4.

6.4 Open problems

Open Problem 1 — VEV coset selection

Select among the 96 $Z_6 \times Z_{16}$-equivalent flavon-VEV cosets on T^2 via string-theoretic R-symmetry or heterotic orbifold fixed-point classification. The leading candidate is a heterotic Z_2^n orbifold assignment consistent with the Z_2-depth matched FN charges (Lean theorem `binaryTreeDepth_matches_FN_charge_magnitude`).

VV coefficients: group-theoretic identities [T] with certified EW mechanism

Layer 1 — VV formula [E]: The log-linear relation is a genuine structural empirical regularity with null density $\leq 10^{-5}$ (SC-VV).
Layer 2 — VV coefficient identities [T]: All three values admit exact SU(5)/SO(10) GUT group-theoretic expressions with zero free parameters:

$$\alpha_d = 1 + \frac{\mathrm{rank}(\mathrm{SU}(5))}{N_c^2} = \frac{13}{9}, \quad \beta_d = -(1 + Y_{Q_L}) = -\frac{7}{6}, \quad \gamma_d = -\frac{\dim(45_{\mathrm{SU}(5)})}{\dim(126_{\mathrm{SO}(10)})} = -\frac{45}{126} = -\frac{5}{14}.$$

The Weyl dimension formula for $A_4 = \mathrm{SU}(5)$ and $D_5 = \mathrm{SO}(10)$ verifies $\dim(45) = 45$ and $\dim(126) = 126$ symbolically; $\gcd(45, 126) = N_c^2 = 9$ cancels. Machine-checked in `ugp-lean` as `VV_from_GUT_group_theory` (zero `sorry`). All three are equivalent to N_c formulas: $\alpha_d = 1 + (N_c{+}1)/N_c^2$, $\beta_d = -(1 + 1/(2N_c))$, $\gamma_d = -(N_c{+}2)/(2(N_c^2{-}2))$ [12].

Honest caveat: These values saturate the pre-registered GUT-representation and discrete-flavor bases (SC-JJJ, SC-LLL null rates $\geq 39\%$; see Sec. 5). The SC-JJJ null test score for the exact GUT-derived values is $\leq 10^{-5}$ (the saturating zone): the structural GUT identities are real but the triple lies in a densely saturated algebraic region.

Layer 3 — EW-scale mechanism [T]: The naive one-loop SM Yukawa RG realization of the log-linear VV functional form fails (anomalous closure; best candidate $\approx 557\%$ vs. $\approx 152\%$ null baseline; artifact `comp_p01_RC1_vv_rg_anomalous_closure.json` [9]). FN-doubled integer-charge frameworks also cannot reproduce VV (hard obstruction; SC-KKK; Sec. 5). The UV/EW-scale mechanism that produces the log-linear relation from the group-theoretic coefficient identities is now identified: the log-linear form arises from composing the SU(5)/SO(10) Yukawa power law with the UCL logarithmic mass map (machine-checked: `vv_mechanism_algebraic`, module `MassRelations.VVMechanism`, zero `sorry`; see [9] for details).

Cross-sector identity (Layer 2 supplement):

$$\dim(126_{\mathrm{SO}(10)}) \;=\; 2 \cdot N_c^2 \cdot \delta \;=\; 2 \cdot 9 \cdot 7 \;=\; 126,$$

where $\delta = 7$ is the charged-lepton mirror offset. Lean: `dim_126_SO10_eq_two_Nc_sq_delta` (zero `sorry`).

Open Problem 3 — Full CKM + CP phase

Extend the Phase-1 per-element 4–6% CKM match to a structural derivation of the full 3×3 CKM matrix including the CP-violating phase δ_{CP}.

Research direction: CKM from A_2 Weyl-chamber basis mismatch. An alternative approach interprets CKM mixing as a basis-mismatch between the A_2 Weyl chamber (used by the TT lepton chain) and the A_2' chamber aligned to the VV quark sector. The geometric phase offset $\Delta\varphi_{\mathrm{VV}} = 14.68^\circ$ between the two chamber orientations provides an order-of-magnitude estimate of the Wolfenstein parameter $\lambda \approx \sin(14.68^\circ) \approx 0.253$ (PDG: $\lambda = 0.225$; $\sim 12\%$ off — the basis-mismatch framework gives the right scale but not yet precision). The quark-lepton complementarity (QLC) relation $\theta_{12}^{\mathrm{CKM}} + \theta_{12}^{\mathrm{PMNS}} \approx 46.5^\circ$ is consistent with a 45° Koide cone angle to **3.2**%, providing independent support for the A_2 basis interpretation (computational artifact `comp_p01_CKM_basis_mismatch.json`).

Improved Wolfenstein estimate from VV coefficient. A structurally tighter estimate uses the VV Lean-certified coefficient $\alpha_d = 13/9$ directly as a power of the primary flavon VEV:

$$\lambda \;\approx\; \varepsilon_1^{\alpha_d} \;=\; \left(e^{-\pi/3}\right)^{13/9} \;=\; e^{-13\pi/27} \;\approx\; 0.2203 \qquad \text{(PDG: 0.2245; 1.9\% off).} \tag{5}$$

This reduces the tension from 12% to 1.9% — a factor of 6.7 improvement. Both inputs are zero-parameter structural constants: $\varepsilon_1 = e^{-\pi/3}$ is the flavon VEV proved to be the global minimum of the $Z_6 \times Z_{16}$ Cartan potential (`fn_vevs_are_potential_minima`); $\alpha_d = 13/9 = 1 + \mathrm{rank}(\mathrm{SU}(5))/N_c^2$ is the VV Lean-certified coefficient (`VV_from_GUT_group_theory`). A null test scanning all rational powers $\varepsilon_1^{p/q}$ for $p, q \in [1, 24]$ confirms that 13/9 is rank #6 by proximity to λ_{PDG} among 576 candidates, but is uniquely the only near-match with a structural Lean-certified origin (artifact `comp_p19_T01_wolfenstein_improved.json`).

The physical interpretation is the following. In the standard Froggatt-Nielsen framework the bare CKM charge difference between generations 1 and 2 is $\Delta a = 1$, giving $|V_{us}|_{\mathrm{bare}} \sim \varepsilon_1^1$. The VV relation implies that down-type quark masses are coupled to up-type masses with weight $\alpha_d = 1 + \mathrm{rank}(\mathrm{SU}(5))/N_c^2 = 1 + 4/9$, rather than with weight 1. This GUT group-theoretic excess $(\alpha_d - 1) = 4/9$ propagates into the CKM diagonalization as an additive correction to the effective ε_1 charge:

$$\Delta a_{\mathrm{eff}} \;=\; \Delta a_{\mathrm{FN}} + (\alpha_d - 1) \;=\; 1 + \tfrac{\mathrm{rank}(\mathrm{SU}(5))}{N_c^2} \;=\; \tfrac{13}{9} \;=\; \alpha_d. \tag{6}$$

The correction applies specifically to the leading (1,2)-generation ε_1 mixing, where the bare FN charge is $\Delta a = 1$. For matrix elements whose bare ε_1 charge is zero (e.g., V_{cb}, which has $\Delta a = 0$), no correction is expected and the Phase-1 FN estimate is unchanged. This specificity is consistent with computational checks (artifact `comp_p19_CDM_cabibbo_mechanism.json`): the formula $|V_{us}| \approx \varepsilon_1^{\alpha_d}$ gives a 1.9% match while applying α_d to zero-charge elements degrades the prediction, confirming the mechanism is Δa-specific.

The algebraic structure of Eqs. (6)–(5) is now fully Lean-certified in the module `UgpLean.MassRelations.CKMMixing` (11 theorems, zero sorry): `cabibbo_effective_charge` proves $\Delta a_{\mathrm{eff}} = \alpha_d$ from the GUT coefficient; `cabibbo_charge_from_GUT` establishes the GUT group-theory origin $\alpha_d = 1 + \mathrm{rank}(\mathrm{SU}(5))/N_c^2$; and `cabibbo_vev_formula` certifies $|V_{us}|_{\mathrm{CDM}} = \varepsilon_1^{\alpha_d}$. The algebraic FN diagonalization bridge is also certified: `fn_vv_correction_additive` proves that $(\alpha_d - 1)$ shifts the bare FN charge *additively*, and `fn_diagonalization_vv_bridge` certifies the log formula $\Delta a_{\mathrm{eff}} \cdot \log \varepsilon_1 = -13\pi/27$ (both zero sorry). The remaining open physical step is the full 2×2 SVD identification, $|(U_{uL}^\dagger U_{dL})_{12}| = \varepsilon_1^{\alpha_d} \cdot (1 + \mathcal{O}(\varepsilon_1^2))$, which requires Weyl–Courant perturbation theory on $Y_d Y_d^\dagger$; once proved, this would elevate Eq. (5) from a [C] structural hypothesis to a [T] Lean-certified result.

The companion paper [11] takes a different structural route to the Wolfenstein parameters, deriving λ directly from the GTE generation and family counts: $\lambda = N_{\mathrm{gen}}^2/(2^{N_{\mathrm{gen}}} N_{\mathrm{fam}}) = 9/40 = 0.22500$, achieving exact agreement with the PDG value (0.0σ) without the VV correction mechanism used here. The two routes are complementary: the approach in this paper provides the cyclotomic-geometric mechanism (VV GUT coefficient) underlying the 1.9% estimate $\varepsilon_1^{\alpha_d} \approx 0.2203$, while [11] identifies the combinatorial formula that closes the remaining gap.

Open Problem 4 — PMNS + δ_{CP}

Construct the PMNS matrix from the FN-doubled charged-lepton sector and anti-doubled ν_R sector; predict δ_{CP} for DUNE Run-1.

7 Conclusions

We have presented two structural mass relations reducing the nine charged-fermion masses to two scalar inputs, and identified their cyclotomic-12 geometric origin:

1. The TT coefficient $\alpha = \pi/6$ is the A_2 Weyl-chamber half-opening angle, Lean-proved (`angle_alpha1_omega1_eq_pi_div_six`). The 2^g structure is Lean-proved via a binary cascade (`cascadeState_eq_TT`). The full TT formula is structurally realised by a two-flavon FN UV completion with VEVs at Lean-proved potential minima (`fn_vevs_are_potential_minima`).

2. The VV formula is a genuine structural regularity (null density $\leq 10^{-5}$). Its coefficient values $(13/9, -7/6, -5/14)$ are structurally derived from SU(5)/SO(10) GUT group theory: $\alpha_d = 1 + \mathrm{rank}(\mathrm{SU}(5))/N_c^2$, $\beta_d = -(1 + Y_{Q_L})$, $\gamma_d = -\dim(45_{\mathrm{SU}(5)})/\dim(126_{\mathrm{SO}(10)})$. Machine-checked: `VV_from_GUT_group_theory` (zero `sorry`). The classification is [E/C].

3. Combined with the Koide closed form [10], all nine charged-fermion masses are determined from two inputs, end-to-end machine-checked in `UgpLean.MassRelations.PhysicalMasses`.

4. The cyclotomic-12 atom family $\{\pi/6, \pi/8, \cos(\pi/12), \sqrt{3}\}$ unifies the TT and Koide structures in a single Lie-geometric framework.

The Lean 4 formalisation (11 modules, zero sorry, standard Mathlib axioms) demonstrates end-to-end machine-auditability of the prediction chain and sets a template for future fermion-mass structural identifications.

Code Availability

All computational artifacts are in the companion repository [13] under `papers/01_SM/canonical_run/`. Key standalone scripts for this paper: `comp_p01_TT_up_lepton_cyclotomic_identity.py` (TT identity verification and null-density test), `comp_p01_VV_down_linked_to_up_lepton.py` (VV identity verification), `comp_p01_XX_gut_structure_search.py` (structural search over GUT representation-theoretic atoms), `comp_p01_EBF_14_vv_rg_flow.py` and `comp_p01_EBF_15_vv_gj_su5_full.py` (one-loop and full GJ/SU(5) RGE tests), `comp_p01_EBF_16_vv_gut_group_theory.py` (exact Weyl-formula verification for $A_4 = \mathrm{SU}(5)$ and $D_5 = \mathrm{SO}(10)$). Each script is independently reproducible with Python 3.9+ and `numpy` (plus `sympy` for the group-theory scripts). Lean formalisation: `ugp-lean` [12] (`lake build` to verify, Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6). See `PROVENANCE.md` and `REPRODUCE.md` in the companion repository for SHA-256 hashes and step-by-step instructions.

Data Availability

All data used are PDG 2022 central values, publicly available at https://pdg.lbl.gov/.

A Complete Lean Module Listing

The following modules in `UgpLean.MassRelations` are relevant to this paper. All build clean under `lake build`; all zero sorry.

```
UgpLean/MassRelations/
  SU3FlavorCartan.lean         -- A2 Weyl-chamber angle pi/6
  BinaryCascade.lean           -- Binary cascade -> TT closed form
  FroggattNielsen.lean         -- FN UV completion; beta = pi/8
  CartanFlavonPotential.lean   -- Z6 x Z16 potential global minima
  Z2OrbifoldDepth.lean         -- Doubled FN charges from Z2-tree depth
  HeavyFermionTower.lean       -- EFT duality: FN = heavy-fermion tower
  UpLeptonCyclotomic.lean      -- TT scaffolding; beta = alpha/C2(SU3)
  DownRational.lean            -- VV rational form; N_c formulas + GUT group theory
  ClebschGordan.lean           -- CG coefficient structural anchors
  CKMMixing.lean               -- CDM: cabibbo_effective_charge, cabibbo_charge_from_GUT,
                               --   cabibbo_vev_formula; zero sorry
  KoideClosedForm.lean         -- Koide m_tau prediction (see companion paper)
  KoideAngle.lean              -- theta = (N_c^2 - 1)/(4 N_c^2) = 2/9 from N_c = 3;
                               -- N_c structural chain (delta, b1, a_top, strand);
                               -- nu seesaw exponent 29/9 (three decompositions);
                               -- dim(126_SO10) = 2 N_c^2 delta cross-identity
  KoideNewtonFlow.lean         -- S3-equivariant Newton flow
  LeptonMassPrediction.lean    -- Full lepton mass pipeline from (m_e, theta)
  ClaimCBridge.lean            -- Formal TT cascade: (pi/6) * 2^g + pi/8
  PhysicalMasses.lean          -- End-to-end capstone (TT + VV + Koide)

UgpLean/ElegantKernel/Unconditional/
  KGenFullClosure.lean         -- k_gen = phi * cos(pi/10); pentagon-hexagon bridge
```

Repository: https://github.com/novaspivack/ugp-lean (build: `lake build`). Zenodo DOI: [12].

B Derivation of the A_2 Weyl-Chamber Angle

We give an explicit two-dimensional construction of eq. (4).

The A_2 root system in the Cartan plane $\mathbb{R}^2$ has simple roots $\alpha_1 = (1,0)$ and $\alpha_2 = (-1/2, \sqrt{3}/2)$ (at 120°). The fundamental weight ω_1 dual to α_1 satisfies $\langle \omega_1, \alpha_1^\vee \rangle = 1$ and $\langle \omega_1, \alpha_2^\vee \rangle = 0$, where $\alpha^\vee = 2\alpha/\langle \alpha, \alpha \rangle$. In Cartesian coordinates: $\omega_1 = (1/2, \sqrt{3}/6)$ (one can verify $\langle \omega_1, \alpha_1 \rangle / |\alpha_1|^2 = 1/2$ and $\langle \omega_1, \alpha_2 \rangle / |\alpha_2|^2 = 0$).

The angle between $\alpha_1 = (1,0)$ and $\omega_1 = (1/2, \sqrt{3}/6)$ is

$$\theta = \arctan\left(\frac{\sqrt{3}/6}{1/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}.$$

The fundamental Weyl chamber is bounded by the half-planes $\langle v, \alpha_i \rangle \geq 0$ for $i = 1, 2$, which in polar coordinates corresponds to angles $\theta \in [0, \pi/3]$. The interior bisector is at $\theta = \pi/6$, i.e., ω_1 lies on the chamber bisector.

This is the content of `angle_alpha1_omega1_eq_pi_div_six` and `omega1_in_weyl_chamber_interior` (both in `UgpLean.MassRelations.SU3FlavorCartan`).

C Pre-Registered Basis Definitions for SC-JJJ, SC-KKK, SC-LLL

SC-JJJ: GUT-representation basis (96 atoms)

Basis SHA-256 (committed before scan):
`644a7f90bc9682db672a8d8920df80a2c481e232c10f44c316a1321b626e8899`.

Atom classes: classical Lie-group ranks $\{\text{rank}(\text{SU}(n))\}_{n=2}^{8}$, $\{\text{rank}(\text{SO}(n))\}_{n=4}^{10}$, $\{\text{rank}(E_n)\}_{n=6}^{8}$, $\text{rank}(F_4)$, $\text{rank}(G_2)$; representation dimensions of SU(3), SU(5), SO(10) including $\dim(45_{\text{SU}(5)}) = 45$, $\dim(126_{\text{SO}(10)}) = 126$; SM hypercharges $Y_Q = 1/6$, $Y_u = 2/3$, $Y_d = -1/3$, $Y_L = -1/2$, $Y_e = -1$, $Y_H = 1/2$; gauge-boson counts $\{1, 3, 8, 4, 12\}$; small rationals $\{1/n\}_{n=1}^{14}$. DL$\leq$ 3: 603,840 expressions.

SC-KKK: FN integer-charge derivation (analytical)

Pre-commit SHA-256 (artifact `comp_p01_KKK_vv_rg_derivation.json`):
`1c4bb21b387519209b912d5bf0576a54b1e469c6ec25ce2eecfd8975be622698`.

Derived analytically: the FN-doubled framework predicts down-sector charges $q_{d,g}^{(1)} = -(7/6)\cdot 2^{g-1}$ and $q_{d,g}^{(2)} = -13/9$, both non-integer. Seven integer-charge candidates tested computationally; all fail.

SC-LLL: Discrete-flavor basis (97 atoms)

Basis SHA-256 (committed before scan):
`dc210d1240f9d8f8d3438fdf0e2443016f2e698f3d4d857a516360fd9861f180`.

Discrete-symmetry groups from [6]: A_4, S_4, T', A_5, $\Delta(27)$, $\Delta(54)$, $\Sigma(168)$, $\Sigma(216)$, $\Sigma(72)$, $\Sigma(36)$, dihedrals D_3–D_8; subgroup indices and representation dimensions; cyclotomic cosines $\cos(2\pi k/n)$ for $n \in \{3, 4, 5, 6, 7, 8\}$. DL$\leq$ 3: 610,421 expressions.

D SHA-256 Artifact Table

Table 5: SHA-256 hashes of all computational artifacts used in this paper. Artifacts are in `papers/01_SM/canonical_run/`.

Artifact	File	SHA-256 (first 16 hex chars)
SC-TT	`comp_p01_TT_....json`	`d8227f9d651b3d95`
SC-VV	`comp_p01_VV_....json`	`10f1bb1e099d444d`
SC-EEE	`comp_p01_EEE_....json`	`120fd2aa2d2270f1`
SC-JJJ	`comp_p01_JJJ_....json`	`f82288b4e6816cfe`
SC-KKK	`comp_p01_KKK_....json`	`e9956111050db7b6`
SC-LLL	`comp_p01_LLL_....json`	`db997f06ff06160f`
SC-BBB	`comp_p01_BBB_....json`	`ef33c48fa938e3fa`

E Six β-Free Inter-Generational Identities

Table 6: The six β-free inter-generational identities derived from the TT formula by taking differences $R_g - R_{g'}$. No free parameters.

Identity	Predicted	PDG	Residual (%)	Note
$\log(m_c/m_e) - \log(m_u/m_e) = (\pi/6)(2^2 - 2^1)$	5.890	5.868	+0.37	
$\log(m_t/m_\tau) - \log(m_c/m_\mu) = (\pi/6)(2^3 - 2^2)$	5.243	5.250	−0.13	
$\log(m_t/m_\tau) - \log(m_u/m_e) = (\pi/6)(2^3 - 2^1)$	11.133	11.118	+0.14	
$\log(m_c m_e/m_u m_\mu)$	$\pi/3$	1.044	−0.03	
$\log(m_t m_\mu/m_c m_\tau)$	$\pi/3$	1.046	−0.16	
$\log(m_t m_e/m_u m_\tau)$	π	3.134	−0.23	Gelfond-type

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[12] Nova Spivack. ugp-lean: Lean 4 formalization of UGP and GTE. https://github.com/novaspivack/ugp-lean, 2026. https://doi.org/10.5281/zenodo.19554700. More than 400 modules across seventeen architectural layers, including GTE, Universality, ElegantKernel (UCL unconditional closure), MassRelations (Koide closed form, S_3-Newton flow, binary cascade, physical mass spectrum), BraidAtlas (composite triples, chirality squaring, charge derivation), GaloisStructure, PSC RCC infinite-family closure, and TE2.2 scan certification. `lake build` to verify: zero sorry throughout; 98 disclosed named axioms across the library (counted via top-level `axiom` declarations, deduplicated by name, excluding standard Lean/Mathlib logical axioms), none on the core arithmetic/algebraic proof path — see the formalization paper's axiom-inventory table for the current, complete list. Lean 4.29.0-rc6 / Mathlib 4.29.0-rc6.

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